

Impact of IgG Fc-Oligosaccharides on Recombinant Monoclonal Antibody Structure, Stability, Safety, and Efficacy

Hongcheng Liu 

Product Characterization, Global Analytical and Pharmaceutical Development, Alexion Pharmaceuticals, New Haven, CT 06510

Christine Nowak

Product Characterization, Global Analytical and Pharmaceutical Development, Alexion Pharmaceuticals, New Haven, CT 06510

Bruce Andrien

Early Stage Analytical Sciences, Global Analytical and Pharmaceutical Development, Alexion Pharmaceuticals, New Haven, CT 06510

Mei Shao

Late Stage Upstream Development, Global Process Development, Alexion Pharmaceuticals, New Haven, CT 06510

Gomathinayagam Ponniah

Product Characterization, Global Analytical and Pharmaceutical Development, Alexion Pharmaceuticals, New Haven, CT 06510

Alyssa Neill

Product Characterization, Global Analytical and Pharmaceutical Development, Alexion Pharmaceuticals, New Haven, CT 06510

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Glycosylation of the conserved asparagine residue in the CH2 domain is the most common posttranslational modification of recombinant monoclonal antibodies. Ideally, a consistent oligosaccharide profile should be maintained from early clinical material to commercial material for the development of recombinant monoclonal therapeutics, though variation in the profile is a typical result of process changes. The risk of oligosaccharide variation posed to further development is required to be thoroughly evaluated based on its impact on antibody structure, stability, efficacy and safety. The variation should be controlled within a range so that there is no detrimental impact on safety and efficacy and thus allowing the use of early phase safety and efficacy data to support project advancement to later phase. This review article focuses on the current scientific understanding of the commonly observed oligosaccharides found in recombinant monoclonal antibodies and their impact on structure, stability and biological functions, which are the basis to evaluate safety and efficacy. It also provides a brief discussion on critical quality attribute (CQA) assessment with regard to oligosaccharides based on the mechanism of action (MOA). © 2017 American Institute of Chemical Engineers Biotechnol. Prog., 33:1173–1181, 2017

Keywords: recombinant monoclonal antibody, oligosaccharides, critical quality attributes, mechanism of action

Introduction

Recombinant monoclonal antibodies are composed of two identical heavy chains and two identical light chains, covalently connected by interchain disulfide bonds and further stabilized by non-covalent interactions. Each heavy chain contains one variable domain (VH), three constant domains (CH1, CH2, and CH3) and a hinge region. Each light chain contains one variable domain (VL) and one constant domain (CL). The entire antibody can also be viewed as three relatively independent globular fragments including two Fab and one Fc. Each Fab fragment is made of four domains: VL, CL, VH, and CH1. Each Fc is made of two CH2 and two CH3 domains. The two Fab fragments and the Fc fragment

are connected by a flexible hinge region. The Fab fragment is responsible for antigen binding. The Fc fragment is responsible for binding to various Fc receptors, including the major ones, Fcγ receptors, the first component of complement (C1q) and the neonatal receptor (FcRn). The specific binding of recombinant monoclonal antibodies to antigens thus blocking antigens from binding to their targets is the mechanism of action (MOA) for some therapeutics. For others, binding to their specific targets and recruiting various Fc receptors to kill the target cells are the MOA. Yet, there are therapeutics where both the antigen binding and receptor binding are MOA. Binding to the neonatal receptor (FcRn) is not involved in MOA of recombinant monoclonal therapeutics; however, it prolongs the half-life of antibodies.

Recombinant monoclonal antibodies are glycosylated with N-linked oligosaccharides attached to the conserved asparagine residue within the consensus sequence in the CH2

Correspondence concerning this article should be addressed to H. Liu at LiuH@alxn.com

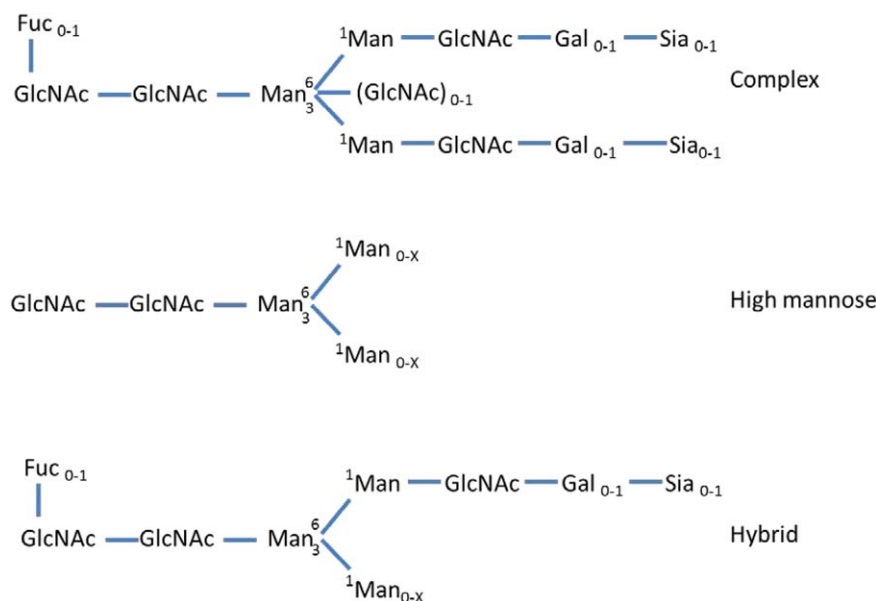


Figure 1. Structures of the three classes of N-linked oligosaccharides. The numbers in subscript after the monosaccharide indicates the absence (0) or presence (1 and X) of the specific monosaccharide. X represents a number that is equal or greater than 1.

domain. Absence of the oligosaccharides causes substantial conformational changes and decreased stability.¹⁻⁷ Without oligosaccharides, the binding of Fc to various receptors and their associated biological functions is either reduced or completely lost.^{1,6,8-10} Studies using various IgG classes indicated that the lack of oligosaccharides either has no impact on half-life^{1,4} or results in shorter in vivo half-life.^{1,11} Human clinical trials did not reveal substantial difference in the half-life of aglycosylated antibodies.¹² Although aglycosylated recombinant monoclonal antibodies have been explored for therapeutic purposes, the majority of recombinant monoclonal antibodies are glycosylated. Glycosylation is one of the main factors that cause recombinant monoclonal antibody heterogeneity. Variation of oligosaccharides is a major topic of discussion for recombinant monoclonal antibody development. Process changes including the use of different cell lines, clones, media type, cell culture conditions, scale, sites and raw materials can all potentially impact the oligosaccharide profiles.^{13,14} The impacts on the product quality, safety and efficacy are required to be thoroughly evaluated. It is also worthwhile to mention that oligosaccharides have been modulated through glycoengineering for optimal therapeutic benefit¹⁵ and for chemical conjugation to make antibody drug conjugates (ADC).¹⁶

The impact of various glycoforms on biological functions has been a topic of several reviews.^{14,17-22} The current review focuses on discussion of the impact of the commonly observed oligosaccharides on antibody structure, stability, and biological functions based on the crystal structures of antibody and antibody in complex with common ligands. Furthermore, we propose that risk assessment of oligosaccharide variation should be based on scientific evidence in literature, mechanisms of action (MOA) of the molecule, knowledge gained during development, and its clinical experience, considering the concept of critical quality attribute (CQA).

Common oligosaccharide structures in recombinant monoclonal antibodies

N-linked oligosaccharides are classified into three categories including complex, high mannose and hybrid (Figure 1).

The three classes of N-linked oligosaccharides share the same core-structure containing two primary N-acetylglucosamines (GlcNAc), one primary mannose (Man) and the two secondary mannose residues. Complex oligosaccharides are composed of GlcNAc and galactose (Gal) and potentially sialic acid on the outer arms. Complex oligosaccharides can exist with or without the core-fucose. High mannose oligosaccharides are composed of Man only in the outer arms. Hybrid oligosaccharides are made of one arm with complex and the other arm with high mannose residues.

In contrast to the complicated oligosaccharide profile with over 30 different glycoforms in human endogenous IgG molecules,^{23,24} only a few structures are present in recombinant monoclonal antibodies expressed in mammalian cell lines with 1-3 major forms accounting for more than 80%, in general.^{25,26} The common types of oligosaccharides in recombinant monoclonal antibodies are shown in Figure 2. The three major ones are complex bi-antennary oligosaccharides with the core-fucose with either zero (G0F), one (G1F) or two (G2F) terminal galactose. To a much lesser extent, high mannose, complex oligosaccharides without the core-fucose, oligosaccharides with sialic acid, and other small forms are also commonly observed in recombinant monoclonal antibodies.

Interactions of oligosaccharides

The conserved asparagine residue for the N-linked glycosylation is located in the CH2 domains, which enclose the oligosaccharides. Because of the bulky oligosaccharide structures, the two CH2 domains have no direct contacts, but only a weak bridge between oligosaccharides from each side.^{27,28} There are extensive interactions between the oligosaccharides and the amino acid residues in the CH2 domains. Oligosaccharides on the 1,6 arm extend in the direction of the CH2-CH3 domain interface and oligosaccharides on the 1,3 arm extend in the direction of the oligosaccharides on the other heavy chain.²⁹ The core structure of GlcNAc2-Man3, GlcNAc on the 1,3 arm and up to the galactose on the 1,6 arm are visible in the crystal structures which

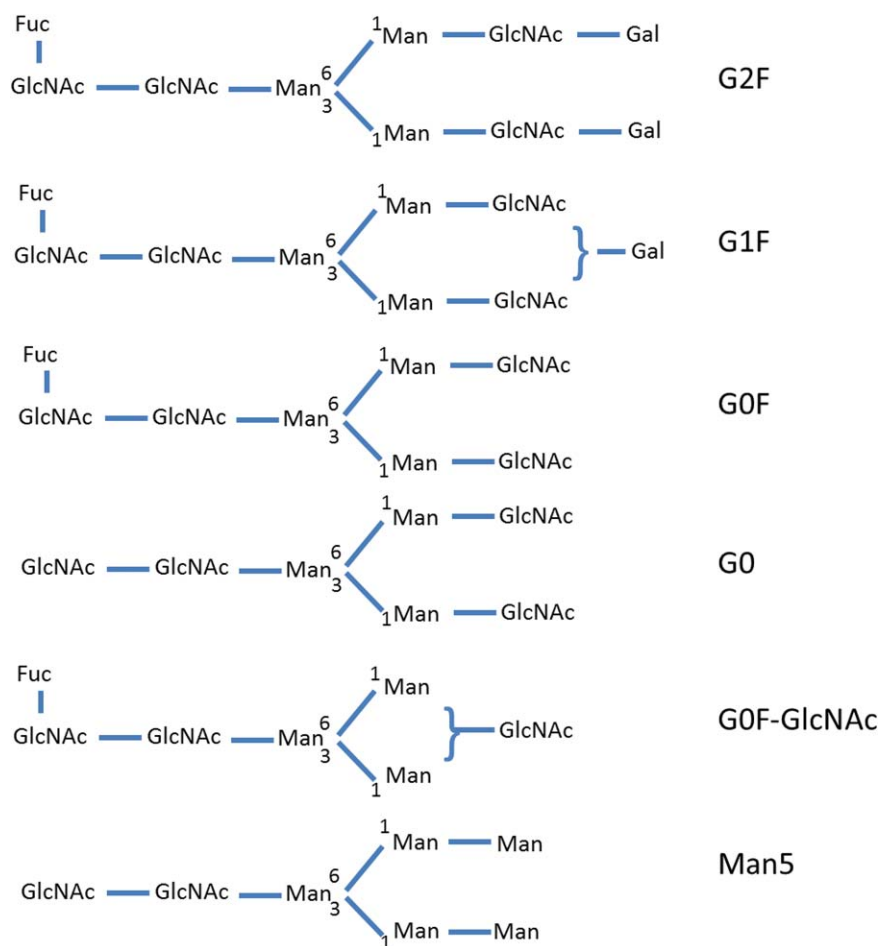


Figure 2. Structures of the commonly observed oligosaccharides of recombinant monoclonal antibodies.

indicates a well-defined structure.^{28,29} The structures are stabilized by interactions between oligosaccharides and amino acids and also interactions between monosaccharides within the carbohydrate structure. Fourteen amino acids are involved in both hydrophobic and hydrogen bonding with the oligosaccharides on the same side of the CH2 domains.²⁸ Oligosaccharides on the 1,3 arm make contact with oligosaccharides on the 1,3-arm of the other heavy chain which is mediated through the mannose residues on the 1,3 arm.²⁹

As implicated from the antibody structure where a flexible hinge separates the Fab and Fc fragments, oligosaccharides have no impact on antigen binding. Oligosaccharides also have no impact on binding to FcRn, as the binding sites are located in the CH2-CH3 domain interface.^{30–32} In contrast, oligosaccharides are critical for the binding of Fcγ receptors and the c1q, which trigger various immune responses. In the crystal structure of an IgG1-Fc in complex with FcγRI, direct contacts only involve the primary GlcNAc of the IgG1-Fc with several amino acids of the receptors.^{33,34} In the crystal structure of human IgG-Fc in complex with FcγRIIa, only two amino acids of the receptor directly interact with the core-fucose of one heavy chain, which plays a minor role in the overall interactions.³⁵ In the crystal structure of IgG-Fc complex with a FcγRIIIb receptor, the first primary GlcNAc of IgG-Fc is involved in direct contact with amino acids of the FcγRIIIb receptor.³⁶ In the crystal structure of IgG-Fc with afucosylated oligosaccharides in complex with FcγRIIIa, the secondary Man on the 1,6-arm also makes a weak contact with the oligosaccharides of the

receptor.³⁷ In the crystal structure of human Fc with receptor RIIIB, no direct contacts between the Fc-oligosaccharides and the receptor were observed.³⁸ There are also no direct contacts between IgG-Fc oligosaccharides with c1q based on the complex crystal structure of c1q and human IgG1.³⁹ Based on the dramatic impact of oligosaccharides on Fcγ receptor and c1q binding and the lack of substantial contact of Fc-oligosaccharides with these ligands, it was proposed that the critical role of oligosaccharides is to maintain and stabilize an IgG-Fc structure that is optimal for binding to various receptors, instead of being directly involved in the binding.^{20,29}

Impact of Various Oligosaccharides on Structure, Stability and Functions

Complex oligosaccharides

Sialic acid. In general, the level of sialic acid in recombinant monoclonal antibodies varies from below 1% to around 15%.^{25,26} The level of sialic acid in human endogenous IgG is around 11%-15%,^{24,26,40} with relatively higher amounts associated with the Fab and lower amounts associated with Fc.⁴¹ The type of sialic acid in human endogenous IgG is N-acetylneuraminic acid (NANA).^{26,40} Recombinant monoclonal antibodies expressed in CHO cell lines contain mainly NANA, while recombinant monoclonal antibodies from murine cell lines contain N-glycolylneuraminic acid (NGNA).^{26,40} NGNA is generally considered to be immunogenic.⁴² It is also worthwhile to mention that sialic acid in

human IgG is in the 2,6-linkage, while IgG from mammalian cell lines are in the 2,3-linkage, although the impact of the linkage on antibody structure and stability is unclear, however, even if there is a difference, its impact is expected to be minimal because of the low percentage and the same chemical nature.

One study demonstrated that there are substantial differences between IgG-Fc with or without the terminal sialic acid as characterized by altered secondary structure, increased exposure of hydrophobic areas, and decreased stability.⁴³ However, most of the studies demonstrate an overall similarity in structure, with only a subtle conformational difference.^{44–49} In a crystal structure, sialic acid on the 1,6 arm, though not in contact with amino acids, can be traced in the electron density map, suggesting a defined conformation. The sialic acids on the 1,6 and 1,3 arms are both solvent exposed.⁴⁴ Conformational changes caused by sialic acid are also evidenced by increased susceptibility of the CH2 domain to limited trypsin^{47,48} and papain⁴⁹ digestion, suggesting increased flexibility around the protease cleavage sites. The observed subtle conformational difference does not cause differences in aggregation propensity under thermal stress, whereas SEC-isolated aggregates and monomer fractions contain the same amount of sialic acid.⁵⁰

Sialic acid associated with the Fc-oligosaccharides has no impact on antigen binding in general.^{8,51,52} However, sialic acid reduces antibody binding to cell surface antigen due to the charge to charge repulsion between the negatively charged sialic acid and the negatively charged cell surface.⁵³ The presence of sialic acid reduced binding to Fcγ receptors and the associated biological activity.⁵¹ A separate study also demonstrated reduced binding to FcγRIIIa and decreased ADCC, which were attributed to decreased binding to Fc receptor as well as decreased binding to cell surface antigens.⁵³ In agreement with the negative impact of sialic acid, increased ADCC was observed for antibodies from cell lines that were co-transfected with sialidase, resulting in antibody with extremely low levels of sialic acid.⁵² In contrast, there are also studies demonstrated that removal of sialic acid has no impact on FcγR1 binding, ADCC and CDC.^{8,54,55} In general, sialic acid has no impact on half-life as demonstrated consistently by several studies.^{51,52,55,56}

Because of the immunogenicity concern, the level of the immunogenic NGNA should be closely monitored and controlled. In addition, sialic acid is a major factor that contributes to the formation of antibody acidic species.^{57,58} Other than that, sialic acid is expected to have minimal impact on the overall product quality and efficacy especially when it is present at such low levels.

Bisecting GlcNAc

Bisecting GlcNAc is very low and only present in antibodies expressed in murine cell lines.^{25,40,59} In human endogenous IgG, bisecting GlcNAc is present at around 10%.²⁴ Bisecting GlcNAc mainly associates with the oligosaccharides in the Fab regions and very little with the oligosaccharides in the Fc region.⁴¹

Bisecting GlcNAc has no impact on antigen binding.⁶⁰ Increased levels of bisecting GlcNAc correlated with increased ADCC,^{60,61} with minimal impact on CDC.⁶⁰ However, this enhancement of ADCC is apparent only when a

relatively high level of bisecting GlcNAc was present in antibodies with mainly core-fucosylated oligosaccharides.⁶²

From the safety point of view, bisecting GlcNAc does not raise any immunogenicity concern because of its natural presence in the human body. Its very low percentage in recombinant monoclonal antibody poses minimal concern for the development of recombinant monoclonal therapeutics.

Galactose. A survey of several commercially available monoclonal antibodies revealed a large variation of the three major glycoforms commonly detected in recombinant monoclonal antibodies, with G0F in the range of approximately 20% to 80%, G1F at 15% to 50% and G2F at 1% to 22%.^{25,26} G0F, G1F, and G2F are also the major glycans in human endogenous IgG with G0F accounting for approximately 20%, G1F at 40%, and G2F at 15% to 20%.^{24,40}

The overall structure of antibodies without the terminal galactose or with various levels of galactose (up to greater than 99%) is very similar.^{3,6,9,63,64} D/H exchange revealed only a subtle local difference around amino acids Leu243, Phe244, and Lys247 with higher levels of protection in the variant with higher level of galactosylation.⁹ An NMR study also indicated a local difference by the observation of a chemical shift around Lys248 and Val308 of an IgG1 after removal of galactose.⁶⁴ The presence of the 100% terminal galactose makes the molecules more prone to digestion compared to IgG without terminal galactose,⁴⁹ suggesting a conformational change around the papain cleavage site. The subtle conformal difference does not impact the aggregation propensity as studies demonstrated that SEC fractions containing either monomer or high molecular weight species after thermal stress⁵⁰ or collected from cell culture⁷ contain the same relative amounts of G0F, G1F, and G2F. Interestingly, slightly higher G2F was found in the large aggregates, but not in the dimer fractions.⁷

The impact of removal of galactose is highly dependent on the IgG subclass, receptor class, and the specific cell types or other factors used for the assays.⁶⁵ There are several studies indicating that removal of terminal galactose has no impact on Fcγ receptor bindings or ADCC.^{6,8,54,55,60,62,63,65–68} Also, several studies have demonstrated that removal of terminal galactose results in a slight decrease in antibody binding to Fcγ receptors.^{9,64,65,69,70} On the other hand, a higher level of galactosylation resulted in a significant increase in binding to this receptor⁹ and higher ADCC.⁷¹ Yet another study demonstrated that removal of galactose from IgG1 and IgG2b decreased their binding to FcγRII, but resulted in increased binding to FcRIII.⁶⁵ The impact of galactose on complement is also complicated as there are studies demonstrating that removal of the terminal galactose has no impact,^{6,54,55} reduced c1q binding^{65,70} and reduced CDC,^{8,60} or slightly increased binding to c1q at lower concentration.⁵⁵ Increased galactosylation correlates with increased ADCC for Rituxan.⁶⁷ The removal of the terminal galactose exposed the high mannose core and allowed the binding of mannose-binding lectin (MBL) leading to complement activation in vitro.^{65,72} The current understanding is that the terminal galactose has minimal impact on ADCC, but more substantial impact on CDC.^{67,73} Galactosylation has no impact on in vivo clearance.^{55,56,74–77}

Overall, the impact of the terminal galactose on antibody structure, stability, and in vivo half-life is minimal. But its impact on ADCC and complement activation may need to be evaluated case-by-case, depending on the mechanisms of

action. The immunogenic α -1,3 galactose^{78,79} that has been commonly detected in recombinant monoclonal antibodies including those expressed in CHO cell lines^{26,26,80} has been a concern from the safety point of view. However, such a concern may just be limited to α -1,3-galactose associated with the Fab-oligosaccharides.⁸¹

Fucose. The majority of the oligosaccharides in recombinant monoclonal antibodies are the complex type with core-fucose. A small percentage of complex oligosaccharides are detected as lacking in core-fucose. In addition, high mannose types of oligosaccharides do not have core-fucose. Similarly, in humans, oligosaccharides with core fucose are the dominant glycan forms, while complex oligosaccharides without the core-fucose and high mannose oligosaccharides are present at much lower levels.^{23,24} The overall structure of antibodies with or without the core-fucose is very similar.^{9,82,83} One study demonstrates that removal of fucose has a subtle impact on the hydration mode of Tyr-296,⁸⁴ supporting the conclusion of similar overall structures between antibodies with or without the core-fucose.

The lack of fucose has no impact on binding of antibodies to Fc γ R1, c1q, FcRn, and the His-polymorphic Fc γ RIIA,^{85,86} but it has slightly increased binding to Arg-polymorphic Fc γ RIIA, and Fc γ RIIB.⁸⁵ The most striking discovery is that antibodies without core fucose showed significantly improved binding to Fc γ RIIA,^{85–88} leading to higher ADCC.^{62,85–87} Increased binding and enhanced ADCC have been shown to increase the efficacy in animal disease models.^{87,89} Interestingly, in this case, stronger interactions between afucosylated IgG to FcRIIA are mainly mediated by carbohydrate-carbohydrate interactions.^{37,90} In one study, afucosylated antibody demonstrated faster clearance in mice,⁸⁷ while a different study did not reveal half-life difference.⁹¹ Because of the therapeutic benefit of afucosylated antibodies, several approaches had been successfully explored to express antibodies without the core-fucose, including the use of cell lines that knockout the gene for the α -1,6 fucosyltransferase,⁸⁶ siRNA,⁹² the use of zinc-finger nucleases that cleave the gene encoding the transferase,⁹³ and depletion of GDP-Fucose by co-expression of GDP-6-deoxy-D-lyxo-4-hexulose reductase.⁹⁴

The center point of discussion with regard to the level of core-fucosylation is its enhanced ADCC, which is beneficial to molecules with the purpose of killing the target cells, but may be detrimental if the receptor function is not wanted. The impact of core-fucose on antibody structure is subtle, therefore, it is not expected to have a significant impact on stability.

Smaller complex oligosaccharides

Several smaller complex oligosaccharides including G0 and G0F-GlcNAc are commonly observed in recombinant monoclonal antibodies.²⁵ These smaller forms are also observed in human endogenous IgG.^{23,24} The percentage of these smaller oligosaccharides is extremely low in both recombinant monoclonal antibodies and in humans.

As discussed in an earlier section, the lack of fucose in G0 is expected to have a subtle local conformational impact. For G0F-GlcNAc, one study demonstrated that loss of the secondary GlcNAc caused a chemical shift of several amino acids around the glycosylation sites measured by NMR.⁶⁴ Loss of the two secondary GlcNAc residues resulted in a

significant loss of thermal stability measured by DSC.^{6,63} Smaller oligosaccharides increased the susceptibility of antibodies to limited trypsin digestion,^{47,48} indicating conformational changes around the glycosylation sites. No difference in aggregation propensity was observed because the same levels of G0, G1 and G0F-GlcNAc were observed in fractions containing monomer or high molecular weight species collected from a recombinant monoclonal antibodies after thermal stress.⁵⁰

Because of the lack of the core-fucose, G0 is expected to have increased binding to Fc γ RIIA and enhanced ADCC. The binding of Fc without the two secondary GlcNAc to soluble Fc γ RIIA was slightly lower compared to antibodies with the two secondary GlcNAc residues.⁶⁴ Removal of the secondary GlcNAc resulted in a slight decrease in binding to Fc γ RII receptors.⁶³ A much lower level of CDC was observed when the two GlcNAc residues were removed.⁶ No differences in half-life of antibodies with G0,^{56,77} G0F-GlcNAc⁷⁶ were observed as compared to antibodies with typical populations of oligosaccharides.

Because of the extremely low level presence, these small complex oligosaccharides are not expected to impact the structure and stability of recombinant monoclonal antibodies. They are also not expected to have any significant impact on antibody safety and efficacy.

High Mannose

The level of high mannose is very low in recombinant monoclonal antibodies, but there are cases in which up to 10% has been observed.^{25,26} In human endogenous IgG, high mannose is present at only around 0.1%.²⁴

D/H exchange revealed that the major structural difference between antibodies with high mannose and complex oligosaccharides is limited to a small region around the glycosylation site.⁴⁸ The structural changes are also evidenced by the observation of increased susceptibility of antibodies with high mannose to proteases.^{47,48} Interestingly, such a small conformational difference resulted in higher resistance to papain digestion.⁸³ The presence of high mannose also decreased antibody thermal stability.⁸³ High mannose has no impact on long term stability⁸³ nor on the propensity towards aggregation by thermal stress.⁵⁰

The presence of high mannose has no impact on antigen binding affinity.^{95,96} Several studies demonstrated that the presence of high mannose reduced binding to Fc γ R1,^{55,95} and Fc γ R2.⁹⁷ On the other hand, one study showed comparable binding affinity of antibody with either high mannose or complex oligosaccharides to Fc γ R1.⁹⁷ The presence of high mannose showed enhanced binding affinity to Fc γ RIIA and enhanced ADCC,^{91,96,97} mainly due to the lack of the core-fucose. Antibodies with high mannose show substantial deficiency in c1q binding and complement activation.^{55,91,95–97} Although high mannose showed deficiency in the classical complement activation, it showed a superior activity through the alternative complement pathway.⁵⁵ There are studies demonstrating no difference in half-life in mice,^{56,75,91,96} but additional studies demonstrated shorter in vivo half-life in humans and animal models.^{55,55,76,77,91,95–97}

The major consideration of high mannose for the development of recombinant monoclonal antibodies should focus on its enhanced ADCC because of the lack of core-fucose. Although antibodies with high mannose showed shorter half-

Table 1. Impact of Oligosaccharides

Oligosaccharides	Safety	Structure	Stability	Fc γ receptors	C1q	Half-life
Sialic acid	NGNA, immunogenicity	Subtle local	No impact	No impact or reduced	No impact	No impact
Bisecting GlcNAc	No impact	Not expected	Not expected	Increased	No impact	Not expected
Galactose	Alfa 1,3 galactose, immunogenicity	Subtle local	No impact	No impact, decreased or increased	Inconclusive: Decreased or slightly increased	No impact
Fucose	No impact	Subtle local	Not expected	Lack of fucose enhancing binding to Fc γ RIIIa	No impact	Inconclusive: No impact or decreased
GO	No impact	Subtle local	Not expected	Increased for Fc γ RIIIa	Not expected	No impact
GOF-GlcNAc	No impact	Subtle local	No impact	Decreased	Decreased	No impact
High mannose	No impact	Subtle local	No impact	Increased for Fc γ RIIIa	Decreased	No impact or decreased
Hybrid	No impact	Subtle local	Not expected	Decreased	Decreased	Decreased

life, this should not have much impact on the half-life of the bulk drug substance due to the low level.

Hybrid

Although more than 10% hybrid has been observed in recombinant monoclonal antibodies,²⁶ in general, its level is extremely low. Human IgG contains around 2% hybrid.²⁴ D/H exchange and limited trypsin digestion demonstrated that a small region (Phe-243 and Y278-F300) with hybrid oligosaccharides is less protected and more flexible in the CH2 domain compared to the same antibody with complex oligosaccharides.⁴⁸ No difference in clearance was observed between antibodies with either complex or hybrid oligosaccharides.⁵⁶

Hybrid oligosaccharides with core-fucose showed decreased binding to FcRIIIa and ADCC compared to IgG with complex oligosaccharides, while antibodies with hybrid oligosaccharides without core-fucose showed stronger binding and enhanced ADCC.⁹¹ Hybrids with or without core-fucose showed no detectable binding to c1q and decreased CDC compared to complex oligos.⁹¹ Hybrid with core-fucose shows faster clearance,⁹¹ probably due to the presence of high mannose on one of the two outer arms

Because of the extreme low levels of hybrid oligosaccharides in recombinant monoclonal antibodies, the presence of hybrid oligosaccharides has not drawn much attention.

CQA Assessment

The impact of various glycan forms on the structure, stability and biological functions are summarized in Table 1. The presence of the same glycoform species at comparable level of each species is critical during various phases of development because it allows the use of safety and efficacy data from earlier phase to extrapolate to the later phase. Variation has to be thoroughly evaluated based on the MOA of the specific molecules. In general, tighter control should be applied to oligosaccharide species that could cause safety and immunogenicity concerns, such as NGNA and α -1,3 galactose. On the other hand, variation in the level of terminal galactose may be more tolerable.

Critical quality attributes (CQA) are defined as "A physical, chemical, biological or microbiological property or characteristic that should be within an appropriate limit, range,

or distribution to ensure the desired product quality."⁹⁸ CQAs are defined based on the evaluation of the severity of an attribute on bioactivity, pharmacokinetics, immunogenicity and safety while taking into account the level of uncertainty.⁹⁹ CQA evaluation with regard to various glycans is highly dependent on the MOA and on the nature of the antigens as being soluble or membrane bound.¹⁰⁰ If MOA is solely by blocking a soluble antigen, glycans play a less critical role. If MOA is dependent on binding to cell surface antigen and utilizing effector functions to kill the target cells, the glycoform is more critical because of its impact on safety and efficacy. In between, if the MOA is binding to cell surface antigen and blocking the binding of the antigen to its target receptors, the major concern of the glycans is safety, where reducing unwanted binding to effector molecules is the goal.

If classified as a CQA, the level of glycoforms is required to be controlled. Even classified as non-CQA, it is still highly desirable to control the levels of various glycans. The main reason is that some of the impact discussed may highly depend on the specific molecules and their associated MOA. Knowledge obtained from one molecule cannot be generalized to be applied to another molecule.

Closing Remarks

Oligosaccharides have been one of the most commonly discussed quality attributes for the development of recombinant monoclonal antibodies. Such discussion, as expected, should focus on safety and efficacy while taking into account the phase of development. The immunogenic oligosaccharides, those with α -1,3 galactose or NGNA, should be avoided, if possible, when considering cell line selection. Their levels should be closely monitored and tightly controlled during the development of monoclonal therapeutic to manage the potential risk. With regard to efficacy, it is less of a concern for antibodies where Fc related functions are not part of the MOA. High mannose, even with shorter in vivo half-life, is not expected to have a substantial impact on bulk drug substance because of the low percentage. On the other hand, if Fc-related functions involve in MOA, it is required to carefully evaluate oligosaccharide variation on efficacy.

Unlike the presence of C-terminal lysine, where strong scientific data and human experience have demonstrated that

there is no impact on safety and efficacy, the scientific evidence of various glycoforms may not yet be strong enough to allow such a generalization. Therefore, the presence of the same types of oligosaccharides at consistent levels should be the goal for the development of recombinant monoclonal therapeutics.

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